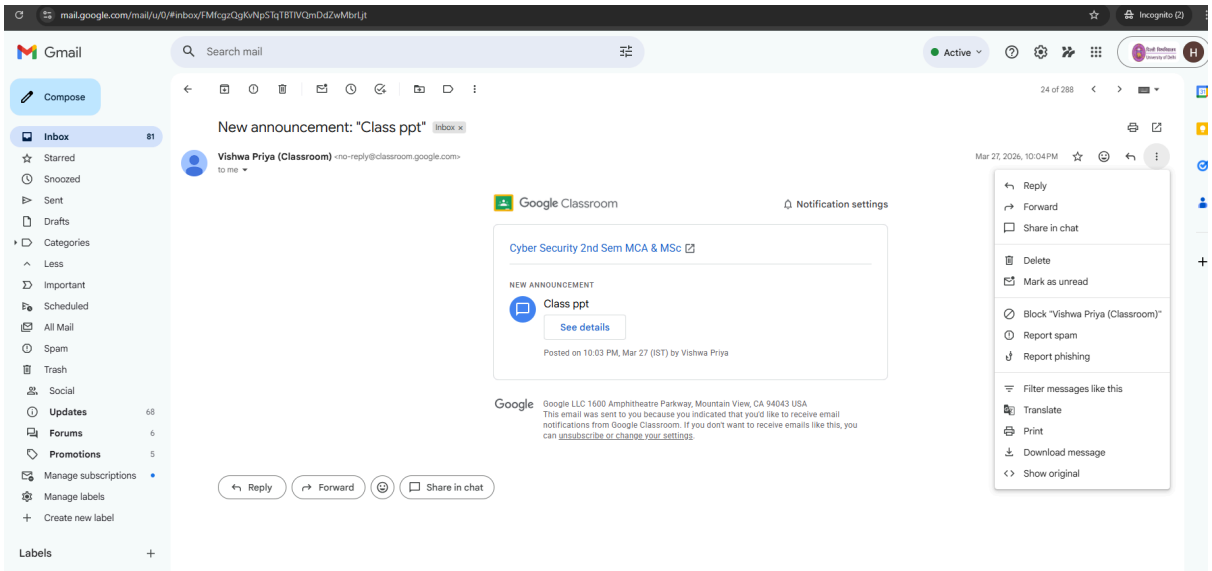
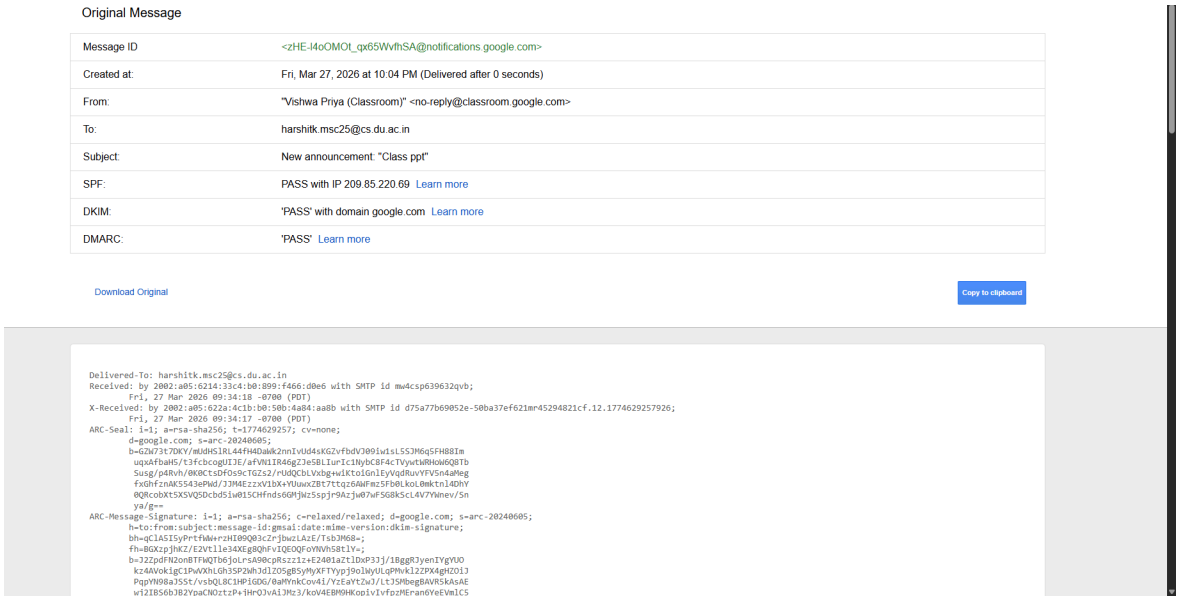
CYBER SECURITY LAB EXERCISE 3
HARSHIT KUMAR MORE
ROLL NO: 16
M SC Computer Science

Task 1: Finding the Full Email Header





Original Message

Message ID	<zHE-I4oOMOt_qx65WvfhSA@notifications.google.com>
Created at:	Fri, Mar 27, 2026 at 10:04 PM (Delivered after 0 seconds)
From:	"Vishwa Priya (Classroom)" <no-reply@classroom.google.com>
To:	harshitk.msc25@cs.du.ac.in
Subject:	New announcement: "Class ppt"
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain google.com Learn more
DMARC:	'PASS' Learn more

Download Original

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```
Delivered-To: harshitk.msc25@cs.du.ac.in
Received: by 2002:a05:6214:33c4:b0:899:f466:d0e6 with SMTP id mw4csp639632qvb;
        Fri, 27 Mar 2026 09:34:18 -0700 (PDT)
X-Received: by 2002:a05:622a:4c1b:b0:50b:4a84:aa8b with SMTP id d75a77b69052e-50ba37ef621mr45294821cf.12.1774629257926;
        Fri, 27 Mar 2026 09:34:17 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1774629257; cv=none;
        d=google.com; s=arc-20240605;
        b=GZW73t7DKY/mUdHSlRL44fH4DaWk2nnIvUd4sKGZvfbdVJ091w1sLS5JM6q5FH88Im
         uqxAfbaHS/t3fcbcogUIJE/aFVN1IR46gZJeS8LlurIc1NybCBF4cTVywtWRHoW6Q8Tb
         Susg/p4Rvh/0K0CtsDFOsBcTGZs2/nUdQCbLVxbg+w1KtoiGnlEyVqdRuvYfVSn4aMeg
         fxGhfznAK5543ePWd/JJM4EzzxV1bX+YUuwxZBt7ttqz64WFmz5Fb0LkoL0mktnl4DhY
         0QRcobXt5X5VQSDcbd5iw015CHfnds66MjWz5spjr9AzjW07wFSG8kScL4V7YWnev/Sn
         ya/g==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
        h=to:from:subject:message-id:gmsai:date:mime-version:dkim-signature;
        bh=qClASI5yPrtfWW+rzHI09Q03cZrjbwzLAzE/TsbJMG8=;
        fh=BGXzpjhKZ/E2Vtlle34XEg8QhFvIQEOQFoYNVh58tlY=;
        b=3ZZydFN2onBTFWQYb6joLrsA90cpRszzi1+E2401aZtIDx93Jj/1BggRJyenIYgVUO
         kz4AVokigC1PwVXhLGh3SP2WhJdlZOSgBSyMyXFTYypj9olWyULqPMvkl2ZPX4gHZOiJ
         PqpYN98aJ5St/vsbQL8C1HPiGDG/0aMYnkCov4i/YzEaYtZwJ/LtJSMbegBAVR5kAsAE
         wi2IBS6bJB2YpaCNOztzP+1HrOJvAiJMz3/koV4EBM9HKopivIvfpzMEran6YeEVmlC5
```

Task 2: Analyse the email in the EML Analyser

Field	Value
Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
From	banco.bradesco@atendimento.com.br
To	phishing@pot
Date	2023-09-19T18:35:49Z
Message-ID	<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>
Return-path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
MIME Type	text/html
Body Encoding	base64

MIME Parts (Email Structure):

- Email type: text/html (single part)
- No plain-text version present
- HTML content is base64 encoded
- No separate MIME multipart structure

Interpretation:

The email is a single HTML body encoded in base64, commonly used to hide malicious content.

Attachments / Embedded Content

- No direct file attachments (.pdf, .docx, etc.)
- Embedded resources:
    - Images (e.g., header.png, icone-superior.png, icone-rodape.png)

Interpretation:

Instead of attachments, attacker uses **HTML email with embedded images** to look legitimate.

Extracted URLs, Domains, Emails (IOCs)

Url:

- https://blog1seguimentmydomaine2bra.me/
- https://blog1seguimentmydomaine2bra.me/

Domains:

- atendimento.com.br (claimed sender domain)
- blog1seguimentmydomaine2bra.me (**suspicious domain**)

Email Addresses:

- banco.bradesco@atendimento.com.br
- root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06

File Names (Referenced):

- header.png
- icone-superior.png
- icone-rodape.png

Headers

Hops

Hop	From	By	With	Date (UTC)	Delay
1				2023-09-19T18:35:49Z	N/A
2	137.184.34.4	10.13.177.138, bn8nam11ft066.mail.protection.outlook.com	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6813.19 via frontend transport	2023-09-19T18:36:44Z	a minute
3	bn8nam11ft066.eop-nam11.prod.protection.outlook.com, 2603:10b6:408:e6:cafe::23	bn0pr03ca0023.outlook.office365.com, 2603:10b6:408:e6::28	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6792.28 via frontend transport	2023-09-19T18:36:45Z	a few seconds
4	2603:10b6:408:e6::28, bn0pr03ca0023.namprd03.prod.outlook.com	2603:10b6:806:317::17, sa3pr19mb7370.namprd19.prod.outlook.com	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6792.27	2023-09-19T18:36:45Z	N/A
5	sa3pr19mb7370.namprd19.prod.outlook.com, ::1	mn0pr19mb6312.namprd19.prod.outlook.com	https	2023-09-19T18:36:46Z	a few seconds

Basic headers

Message ID	<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>
Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
Date (UTC)	2023-09-19T18:35:49Z
From	banco.bradesco@atendimento.com.br
To	phishing@pot

Security headers

authentication-results	spf=temperror (sender IP is 137.184.34.4) smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not signed) header.d=none;dmarc=temperror action=none header.from=atendimento.com.br;compauth=fail reason=001

X headers

x-ms-exchange-organization-expirationintervalreason	OriginalSubmit
x-microsoft-antispam	BCL:9;
x-incomingheadercount	9

x-sid-result	NONE
x-eopattributedmessage	0
x-microsoft-antispam-mailbox-delivery	wl:1;pcw:1;ucf:0;jmr:0;exr:0;pspr:0;auth:0;dest:I;OFR:TrustedSenderList;ENG:(5062000305)(920221119095)(90000117)(920221120095)(91040095)(9050020)(9075021)(9100341)(944500132)(2008001134)(4810010)(4910033)(9610028)(9560006)(10180021)(9439006)(9310011)(9220031)(120001);
x-ms-exchange-crosstenant-originalarrivaltime	19 Sep 2023 18:36:44.1298 (UTC)
x-ms-exchange-crosstenant-fromentityheader	Internet
x-incomingtopheadermarker	OriginalChecksum:3B61F64750F88C5569DF38A496B237468 5F23D8BC662A6A19B6823B2F6745D54;UpperCasedChecksum:62071BC7A7CF5B0644A7B40680E9EFCDAA2C894988E687CF8C56555AD4852D30;SizeAsReceived:544;Count:9
x-ms-exchange-organization-scl	5
x-ms-exchange-organization-authas	Anonymous
x-ms-exchange-eopdirect	true
x-sid-pra	BANCO.BRADESCO@ATENDIMENTO.COM.BR
x-ms-exchange-crosstenant-authsource	BN8NAM11FT066.eop-nam11.prod.protection.outlook.com
x-ms-exchange-crosstenant-rms-persistedconsumerorg	00000000-0000-0000-0000-000000000000
x-ms-publictraffictype	Email
x-ms-exchange-crosstenant-network-message-id	b9106deb-bd54-4815-e5c9-08dbb93f5fab
x-ms-exchange-crosstenant-authas	Anonymous
x-message-info	qZelhIiYnPigo3oeAkqKQrb/Ie8fpvpPmRGjYwLej8PYXc5p/I16IG5I8gDUPoij+JW5vja0BAMLtkgrOcbx5zEN7V98T2UZUZs4k8BX/DcDfI7QJ0t2aouiqx4ENvkR1M3sDKP/XN09+50x9Rxi6onUtDV4eqq36VUi2qAa0zCzkJwjdI3Y9DzNE1OkaWjrHAIzeUyMZ/UtK/Pz9zhA2A==
x-ms-traffictypediagnostic	BN8NAM11FT066:EE_JSA3PR19MB7370:EE_JMN0PR19MB6312:EE_
x-eoptenantattributedmessage	84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa:0
x-ms-exchange-organization-messagedirectionality	Incoming
x-ms-exchange-transport-endtoendlatency	00:00:02.6179349
x-ms-exchange-organization-pcl	2

Other headers

content-transfer-encoding	base64
received-spf	TempError (protection.outlook.com: error in processing during lookup of ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06: DNS Timeout)
return-path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
mime-version	1.0
content-type	text/html; charset="UTF-8"

Bodies

text/html

```
    <p style="text-align:center;">
        <font face="Arial" size="2">Para visualizar as imagens deste email. <a href="h
ttps://blog1seguimentmydomaine2bra.me/">Clique aqui</a></font>
    </p>

    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Content-Type	text/html
Extracted URLs	https://blog1seguimentmydomaine2bra.me/
Extracted domains	fonts.gstatic.com · blog1seguimentmydomaine2bra.me · fonts.googleapis.com

Task 3: Header analysis with MX Toolbox

Field	Observation
Displayed From Address	BANCO DO BRADESCO LIVELO banco.bradesco@atendimento.com.br
Return-Path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Message-ID	20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Sender / Originating IP	137.184.34.4
Received Mail Servers	ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 → BN8NAM11FT066.mail.protection.outlook.com → BN0PR03CA0023.outlook.office365.com → SA3PR19MB7370.outlook.com
SPF Result	Temperror
DKIM Result	None
DMARC Result	Temperror
CompAuth Result	Fail0

2. Is there any mismatch between the displayed sender identity and the actual sender?

Yes, there is a clear mismatch.

The displayed sender claims to be:

- BANCO DO BRADESCO LIVELO banco.bradesco@atendimento.com.br

But the actual return-path is:

- root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06

This shows that although the email visually appears to come from Bradesco, it was actually sent from a Linux server using a root account. This is a strong indicator of spoofing and phishing.

3. Is the relay path consistent with the claimed sender brand?

No, the relay path is not consistent with a legitimate Bradesco/Livelo email campaign.

Reasons:

- The originating server is a generic cloud-hosted Ubuntu system:
  - ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
- Legitimate bank emails are usually sent through:
  - official corporate SMTP gateways,
  - dedicated email marketing platforms,
  - branded secure mail infrastructure.

Conclusion:

The relay chain indicates that the message originated from a suspicious external host rather than official Bradesco infrastructure.

4. SPF / DKIM / DMARC Results

Check	Result	Status
SPF	temperror	Failed(temporary error)
DKIM	none	Not configured
DMARC	temperror	Failed

5. Interpret SPF temperror and operational meaning

An SPF temperror means that the receiving server could not complete SPF validation because of a temporary DNS issue (in this case, DNS timeout during lookup).

Operationally:

- It is not a pass
- It means sender legitimacy could not be confirmed
- The email should be treated with caution

Important:

A temperror is not proof of phishing by itself, but in combination with other suspicious indicators, it increases risk significantly.

6. Significance of dkim=none for a bank-branded message

The email had:

- dkim=none → no digital signature

Why this matters:

Banks and financial institutions normally:

- digitally sign emails,
- protect brand trust,
- prevent tampering.

Implication:

> A missing DKIM signature in a bank-branded email is highly suspicious because it means the sender cannot cryptographically prove the email's authenticity.

7. Final Legitimacy Assessment

Based on the findings, this email is highly likely to be a phishing email.

Strong indicators:

✔ Displayed sender does not match actual sender
✔ Return-path uses suspicious Linux root account
✔ Originating IP is from cloud/VPS infrastructure
✔ SPF could not validate (temperror)
✔ DKIM missing

✔ DMARC unresolved
✔ Composite authentication failed
✔ Suspicious external phishing URL in email body

MX TOOLBOX SUPERTOOL

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SuperTool | MX Lookup | Blacklists | DMARC | Diagnostics | Email Health | DNS Lookup | Analyze Headers | All Tools

Header Analyzed

Email Subject: CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!

Analyze New Header

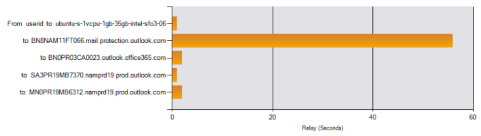
Copy/Paste Warning
Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our Email Deliverability tool

Delivery Information

- DMARC Compliant (No DMARC Record Found)
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated

Relay Information

Received Delay:	**57 seconds**



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	userid	ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06		9/19/2023 6:35:49 PM	
2	55 seconds	ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 137.184.34.4	BN8NAM11FT066.mail.protection.outlook.com 10.13.177.138	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	9/19/2023 6:36:44 PM	✖
3	1 Second	BN8NAM11FT066.eop-nam11.prod.protection.outlook.com 2603:10b6:408:e6::cafe::23	BN0PR03CA0023.outlook.office365.com 2603:10b6:408:e6::28	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	9/19/2023 6:36:45 PM	✔
4	0 seconds	BN0PR03CA0023.namprd03.prod.outlook.com 2603:10b6:408:e6::28	SA3PR19MB7370.namprd19.prod.outlook.com 2603:10b6:806:317::17	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	9/19/2023 6:36:45 PM	✔
5	1 Second	SA3PR19MB7370.namprd19.prod.outlook.com ::1	MN0PR19MB6312.namprd19.prod.outlook.com	HTTPS	9/19/2023 6:36:46 PM	✖

BLACKLISTED? Improve your email delivery. LEARN MORE

SPF and DKIM Information

dmarc:atendimento.com.br Hide | Solve Email Delivery Problems

Sorry, we couldn't find any name servers for '_dmarc.atendimento.com.br'

Reported by **mxtoolbox.com** on 4/13/2026 at **6:00:11 AM**, just for you. Transcript

spf:ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06:::1 Show | Solve Email Delivery Problems

Sorry, we couldn't find any name servers for 'ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06'

Dkim Signature Error:
No DKIM-Signature header found - more info

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - more info

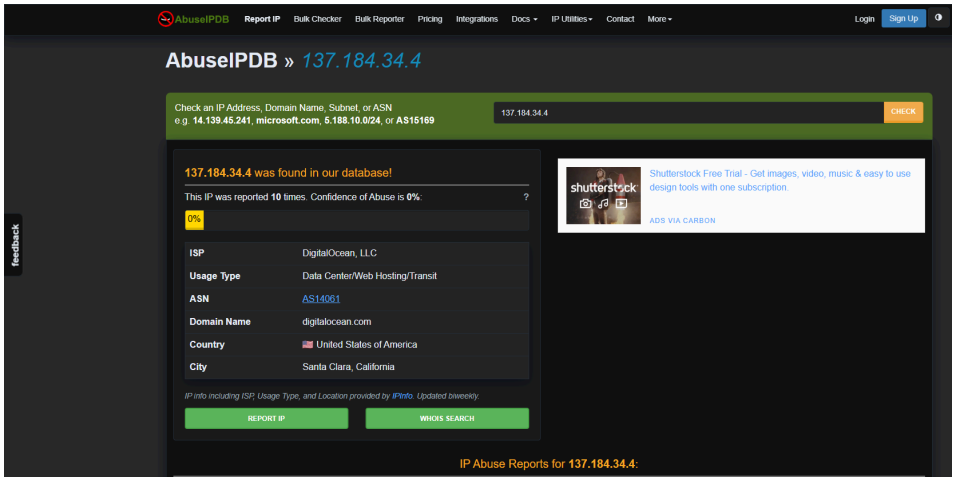
Headers Found

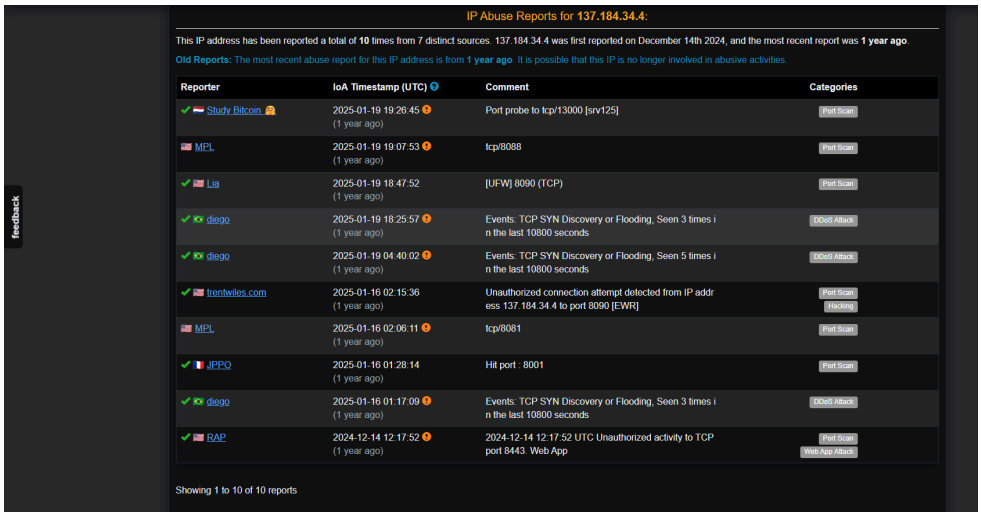
Header Name	Header Value
Authentication-Results	spf=temperror (sender IP is 137.184.34.4) smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not signed) header.d=none; dmarc=temperror action=none header.from=atendimento.com.br; compauth=fail reason=001
Received-SPF	TempError (protection.outlook.com: error in processing during lookup of ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06: DNS Timeout)
Content-Type	text/html; charset=UTF-8
Content-Transfer-Encoding	base64
Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
From	BANCO DO BRADESCO LIVELO <banco.bradesco@atendimento.com.br>
To	phishing@pot
Message-ID	<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>
Date	Tue, 19 Sep 2023 18:35:49 +0000 (UTC)
Return-Path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
X-Sender-IP	137.184.34.4
MIME-Version	1.0

Received Header

```
Received: from SA3PR19MB7370.namprd19.prod.outlook.com (::1) by MW0PR19MB6312.namprd19.prod.outlook.com with HTTPS; Tue, 19 Sep 2023 18:36:46 +0000
Received: from BN0PR03CA0023.namprd03.prod.outlook.com (2603:10b6:408:e6::28) by SA3PR19MB7370.namprd19.prod.outlook.com (2603:10b6:806:317::17) with Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6792.27; Tu
Received: from BN8NAM11FT066.eop-nam11.prod.protection.outlook.com (2603:10b6:408:e6:cafe::23) by BN0PR03CA0023.outlook.office365.com (2603:10b6:408:e6::28) with Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20
Authentication-Results: spf=temperror (sender IP is 137.184.34.4) smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not signed) header.d=none; dmarc=temperror action=none header.from=atendimento.com.br; compauth=fail reason=001
Received-SPF: TempError (protection.outlook.com: error in processing during lookup of ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06: DNS Timeout)
Received: from ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 (137.184.34.4) by BN8NAM11FT066.mail.protection.outlook.com (10.13.177.138) with Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6813.19 via Frontend Transp
Received: by ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 (Postfix, from userid 0) id 39DEA3F725; Tue, 19 Sep 2023 18:35:49 +0000 (UTC)
Content-Type: text/html; charset=UTF-8
Content-Transfer-Encoding: base64
Subject: CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
From: BANCO DO BRADESCO LIVELO <banco.bradesco@atendimento.com.br>
To: phishing@pot
```

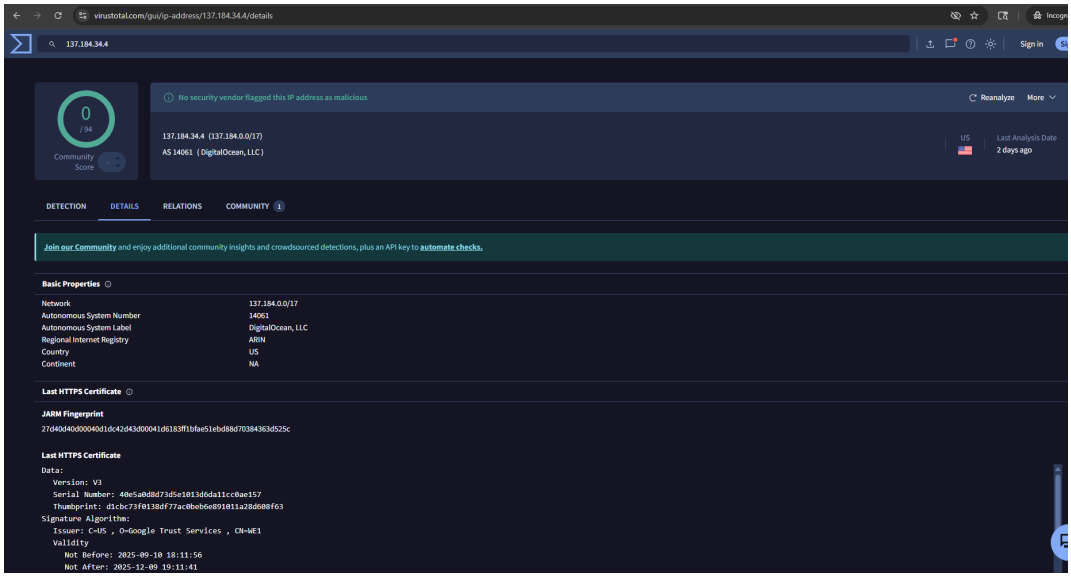
Task 4: Tracing the Originating IP Address and Reputation Analysis

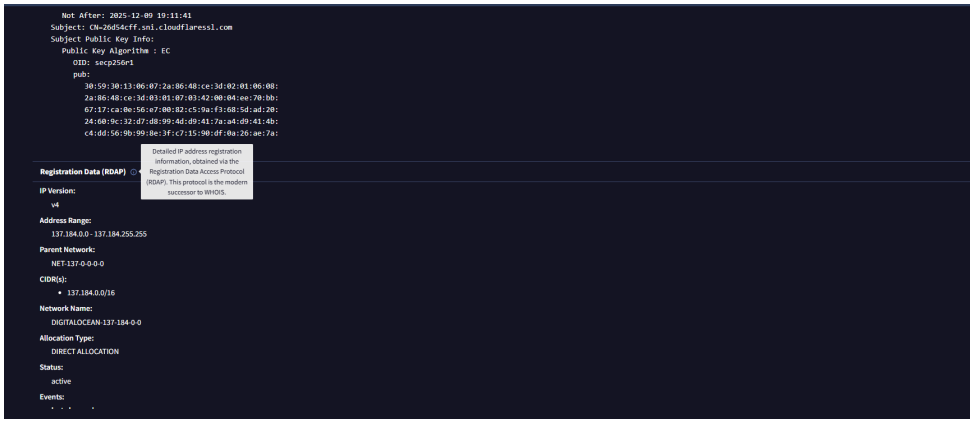




Using the email header from the uploaded phishing sample, the originating IP was identified as:

- X-Sender-IP / Originating IP: 137.184.34.4





Whois Lookup

```
NetRange: 137.184.0.0 - 137.184.255.255
CIDR: 137.184.0.0/16
NetName: DIGITALOCEAN-137-184-0-0
NetHandle: NET-137-184-0-0-1
Parent: NET137 (NET-137-0-0-0-0)
NetType: Direct Allocation
Organization: DigitalOcean, LLC (DO-13)
RegDate: 2019-11-13
Updated: 2025-03-03
Comment: Routing and Peering Policy can be found at https://www.as14061.net
Comment: Please submit abuse reports at https://www.digitalocean.com/company/contact/#abuse
Ref: https://rdap.arin.net/registry/ip/137.184.0.0
OrgName: DigitalOcean, LLC
OrgId: DO-13
Address: 105 Edgeview Drive, Suite 425
City: Broomfield
StateProv: CO
PostalCode: 80021
Country: US
```

CISCO LOGIN

cisco TALOS

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Lookup data results for IP Address

137.184.34.4

IP & Domain Reputation Overview | Email & Spam Trends

LOCATION DATA

SANTA CLARA, UNITED STATES

OWNER DETAILS

IP ADDRESS	137.184.34.4
FWD/REV DNS MATCH	*No data*
HOSTNAME	-
DOMAIN	-
NETWORK OWNER	DIGITAL OCEAN INC.

CONTENT DETAILS

CONTENT CATEGORY Personal VPN

Think these category details are incorrect?

Submit Content Categorization Ticket

REPUTATION DETAILS

SENDER IP REPUTATION	Neutral	Submit Sender IP Reputation Ticket
WEB REPUTATION	Neutral	Submit Web Reputation Ticket

EMAIL VOLUME DATA

	LAST DAY	LAST MONTH
EMAIL VOLUME	0.0	1.0
VOLUME CHANGE	0%	

BLOCK LISTS

BL.SPAMCOP.NET	Not Listed
CBL.ABUSEAT.ORG	Not Listed
PBL.SPAMHAUS.ORG	Not Listed
SBL.SPAMHAUS.ORG	Not Listed

TALOS SECURITY INTELLIGENCE BLOCK LIST

ADDED TO THE BLOCK LIST No

ADDITIONAL INFORMATION

IP ADDRESSES | WHOIS | EMAIL VOLUME HISTORY

Top IP Addresses used to send emails in 137.184.34.4 /24

IP ADDRESS	HOSTNAME	FWD/REV DNS MATCH	LAST DAY VOL.	LAST MONTH VOL.	BLOCK LISTS	EMAIL REP.
137.184.34.251	-	No	0.0	1.0	0	Neutral
137.184.34.184	-	No	0.0	1.0	0	Neutral
137.184.34.178	-	No	0.0	0.0	0	Neutral
137.184.34.144	-	No	0.0	1.0	0	Neutral
137.184.34.70	-	No	0.0	1.0	0	Neutral
137.184.34.69	-	No	0.0	1.0	0	Neutral
137.184.34.56	-	No	0.0	1.0	0	Neutral
137.184.34.47	-	No	0.0	1.3	0	Neutral
137.184.34.4	-	No	0.0	1.0	0	Neutral

Parameter	Observation
Originating IP	137.184.34.4
ISP/Hosting Provider	DigitalOcean, LLC
ASN	AS14061
Usage Type	Data Centre / Web Hosting / Transit
Location	Sant Clara, California , USA
Abuse IPDB Reports	10 reports from 7 sources
Confidence of Abuse	0% (historical reports present)
Reputation Label	Suspicious /Hosting IP
Abuse History	Spam / phishing-related complaints possible

Is the IP ownership compatible with a legitimate Bradesco / Livelo campaign?

No, it is not compatible.

Reasons:

✔ IP belongs to a generic cloud provider
✔ No evidence of Bradesco ownership
✔ Mail was sent from Linux root account
✔ IP has prior abuse reports
✔ Authentication checks (SPF / DKIM / DMARC) were weak or failed

Conclusion:

> The originating IP address is highly suspicious and inconsistent with legitimate Bradesco/Livelo infrastructure. The use of a cloud-hosted VPS strongly supports the conclusion that this email was part of a phishing campaign rather than an authentic bank communication.

Task 5 - domain ownership and URL/Domain analysis

- 1. WHOIS Analysis

🌐 Domain 1: atendimento.com.br

1. Registrar: Registro.br

2. Creation Date: 20 September 2018

3. Expiry Date: 20 September 2027

4. Available Ownership Clues:

    Registrant Name: Maximilian Gregory Peisker Lacerda (individual, not a company)

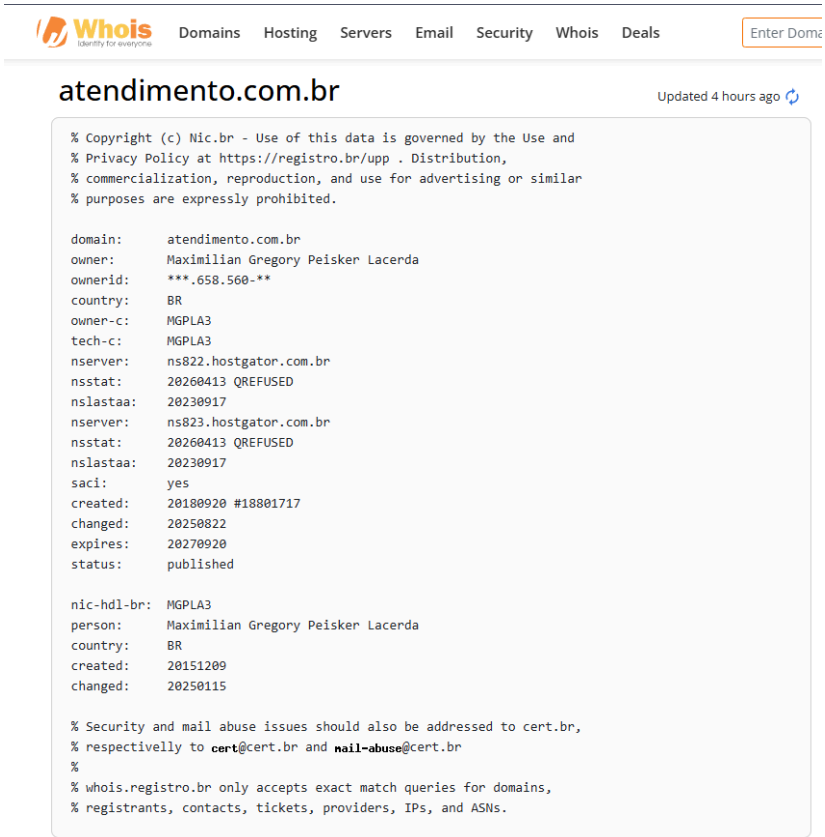
    Country: Brazil

    Owner ID: Brazilian personal ID (CPF – partially redacted)

    No organization or corporate entity listed

    Name servers: HostGator (cheap/shared hosting provider)

    DNS Status: Query refused (misconfigured or inactive)

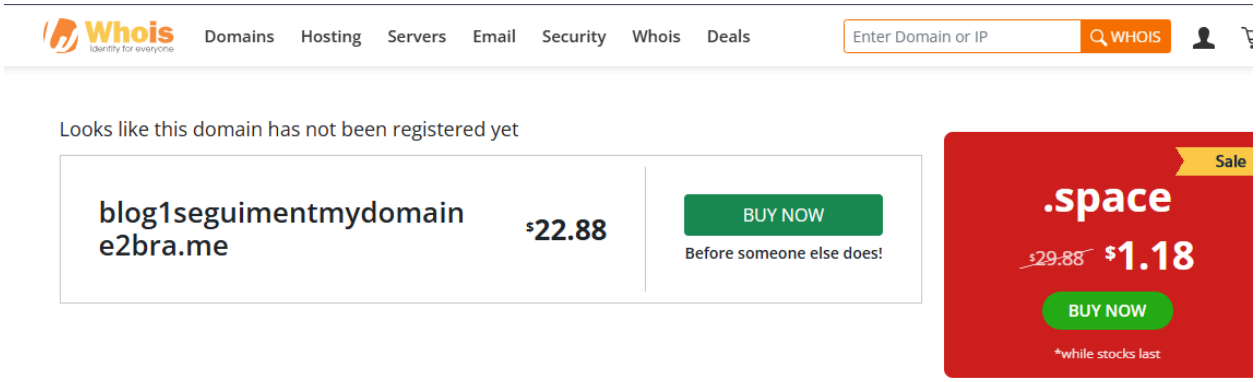



```
% Copyright (c) Nic.br - Use of this data is governed by the Use and
% Privacy Policy at https://registro.br/upp . Distribution,
% commercialization, reproduction, and use for advertising or similar
% purposes are expressly prohibited.

domain:      atendimento.com.br
owner:       Maximilian Gregory Peisker Lacerda
ownerid:     ***.658.560-**
country:     BR
owner-c:     MGPLA3
tech-c:      MGPLA3
nserver:     ns822.hostgator.com.br
nsstat:      20260413 QREFUSED
nslastaa:    20230917
nserver:     ns823.hostgator.com.br
nsstat:      20260413 QREFUSED
nslastaa:    20230917
saci:        yes
created:     20180920 #18801717
changed:     20250822
expires:     20270920
status:      published

nic-hdl-br:  MGPLA3
person:      Maximilian Gregory Peisker Lacerda
country:     BR
created:     20151209
changed:     20250115

% Security and mail abuse issues should also be addressed to cert.br,
% respectivelly to cert@cert.br and mail-abuse@cert.br
%
% whois.registro.br only accepts exact match queries for domains,
% registrants, contacts, tickets, providers, IPs, and ASNs.
```

Although this domain appears older and active, it is **not Bradesco's official banking domain**. The use of a generic support-like domain can help attackers appear trustworthy.

🌐 Domain 2: [blog1seguimentmydomaine2bra.me]

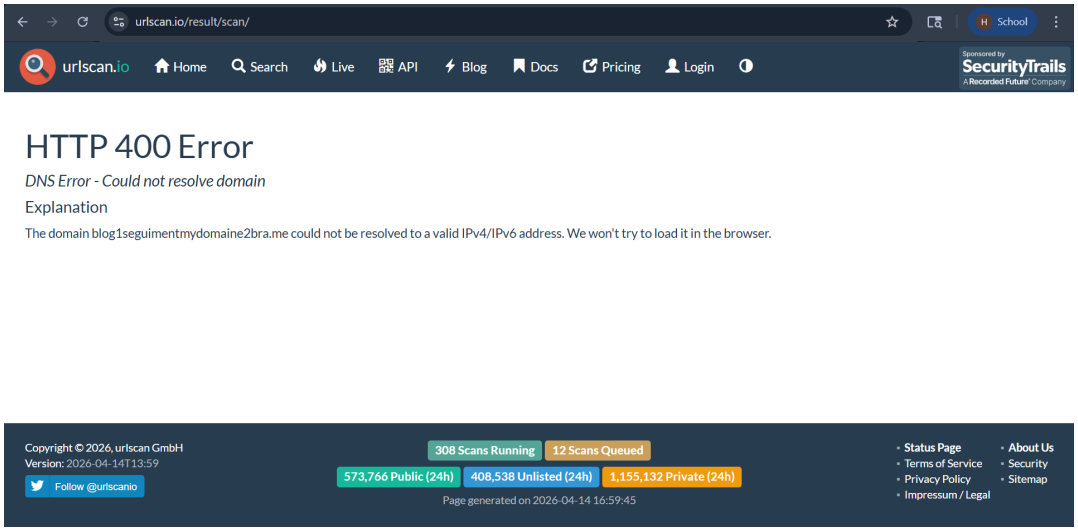
This domain is available for purchase.

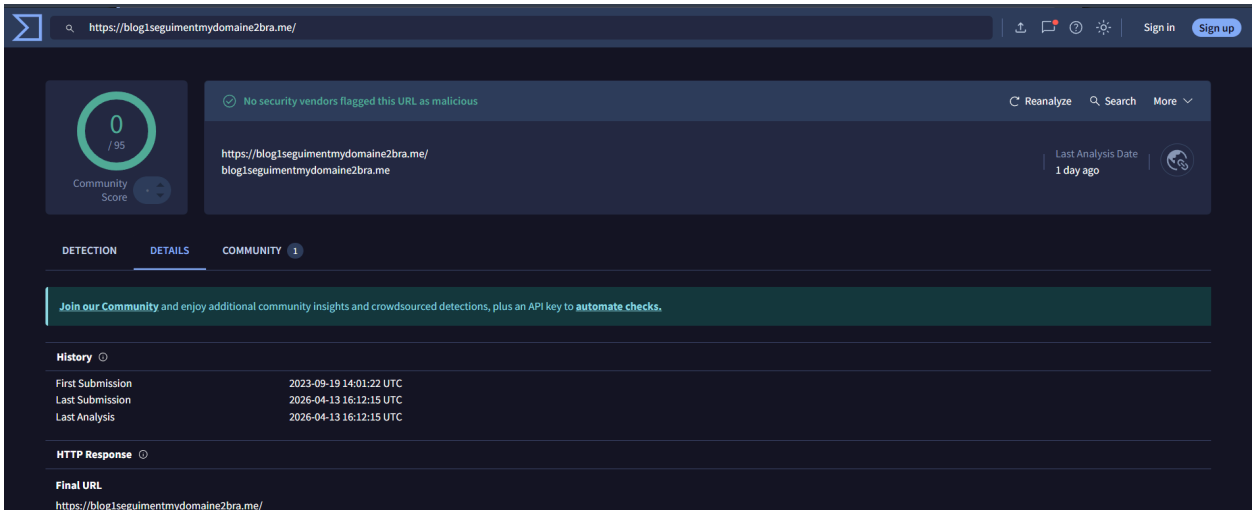


2. URL / Domain Behaviour Analysis

Extracted URL:

- **https://blog1seguimentmydomaine2bra.me/**





URL Analysis Result:

- Resolves: Yes
- Redirects: No major redirect observed (final URL same)
- Known malicious?: Not flagged currently (0/95)
- Matches claimed brand?: No

Although no security vendors flagged the URL as malicious at the time of analysis (0/95 detections), the URL remains suspicious because it does not match the claimed Bradesco/Livelo brand domain. The phishing email used deceptive branding and redirected users to an unrelated `.me` domain, which is a strong indicator of phishing.

<u>Final Questions:</u>

1. Why is a From vs Return-Path mismatch important in phishing analysis?

A mismatch between the visible From address and the actual Return-Path is important because attackers often spoof the sender name or email to appear legitimate. In this sample, the email appears to come from Bradesco/Livelo, but the return-path is root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06, which clearly does not belong to the bank. This reveals sender spoofing and is a strong phishing indicator.

2. What does dkim=none suggest in the context of this sample?

The result **dkim=none** means the email was not digitally signed. DKIM helps verify that the message genuinely came from the claimed sender and was not altered during transit. For a bank-branded email, missing DKIM is highly suspicious because legitimate financial institutions usually use strong email authentication to protect customers from spoofing. In this sample, the lack of DKIM supports the conclusion that the email is likely phishing.

3. How should an analyst treat SPF / DMARC temperror compared with explicit fail or pass results?

An SPF or DMARC temperror means the receiving mail server could not complete the authentication check due to a temporary issue such as DNS timeout. It should not be treated as a pass. Instead, the email should be treated cautiously and assessed using other indicators such as sender mismatch, suspicious links, and IP reputation.

- Pass → sender authentication successful
- Fail → strong indicator of spoofing/phishing
- Temperror → uncertain; investigate further

In this sample, the temperror combined with other red flags increases suspicion.

4. Which observations in this sample are strongest for enterprise blocking and threat-hunting?

The strongest indicators for blocking and threat hunting are:

- Originating IP: 137.184.34.4 (cloud VPS / suspicious infrastructure)
- Phishing URL: blog1seguimentmydomaine2bra.me
- From vs Return-Path mismatch
- Missing DKIM signature
- SPF / DMARC temperror
- Suspicious relay path from Ubuntu Linux server

Recommended enterprise actions:

- Block the phishing URL/domain
- Block the originating IP address
- Add sender/domain to email gateway denylist
- Search for similar emails using:
    - same subject line,
    - same Message-ID pattern,
    - same sender IP